

C1 Equation of a straight line

Name: _____

Class: _____

Date: _____

Time: **446 minutes**

Marks: **372 marks**

Comments:

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Q1.

A circle with centre C has equation $x^2 + y^2 - 6x + 10y + 9 = 0$.

- (a) Express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(3)

- (b) Write down:

- (i) the coordinates of C ;
- (ii) the radius of the circle.

(2)

- (c) The point D has coordinates $(7, -2)$.

- (i) Verify that the point D lies on the circle.

(1)

- (ii) Find an equation of the normal to the circle at the point D , giving your answer in the form $mx + ny = p$, where m , n and p are integers.

(3)

- (d) (i) A line has equation $y = kx$. Show that the x -coordinates of any points of intersection of the line and the circle satisfy the equation

$$(k^2 + 1)x^2 + 2(5k - 3)x + 9 = 0$$

(2)

- (ii) Find the values of k for which the equation

$$(k^2 + 1)x^2 + 2(5k - 3)x + 9 = 0$$

has equal roots.

(5)

- (iii) Describe the geometrical relationship between the line and the circle when k takes either of the values found in part (d)(ii).

(1)

(Total 17 marks)

Q2.

The line AB has equation $3x + 5y = 11$.

- (a) (i) Find the gradient of AB .

(2)

- (ii) The point A has coordinates $(2, 1)$. Find an equation of the line which passes through the point A and which is perpendicular to AB .

(3)

- (b) The line AB intersects the line with equation $2x + 3y = 8$ at the point C . Find the coordinates of C .

(3)

(Total 8 marks)

Q3.

A circle with centre C has equation

$$(x - 5)^2 + (y + 12)^2 = 169$$

- (a) Write down:

- (i) the coordinates of C ;

(1)

- (ii) the radius of the circle.

(1)

- (b) (i) Verify that the circle passes through the origin O .

(1)

- (ii) Given that the circle also passes through the points $(10, 0)$ and $(0, p)$, sketch the circle and find the value of p .

(3)

- (c) The point $A(-7, -7)$ lies on the circle.

- (i) Find the gradient of AC .

(2)

- (ii) Hence find an equation of the tangent to the circle at the point A , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

(Total 11 marks)

Q4.

A curve is defined by the parametric equations

$$x = \frac{1}{t}, \quad y = t + \frac{1}{2t}$$

- (a) Find $\frac{dy}{dx}$ in terms of t .

(4)

- (b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

- (c) Show that the cartesian equation of the curve can be written in the form

$$x^2 - 2xy + k = 0$$

where k is an integer.

(3)

(Total 11 marks)

Q5.

The triangle ABC has vertices $A(-2, 3)$, $B(4, 1)$ and $C(2, -5)$.

- (a) Find the coordinates of the mid-point of BC .

(2)

- (b) (i) Find the gradient of AB , in its simplest form.

(2)

- (ii) Hence find an equation of the line AB , giving your answer in the form $x + qy = r$, where q and r are integers.

(2)

- (iii) Find an equation of the line passing through C which is parallel to AB .

(2)

- (c) Prove that angle ABC is a right angle.

(3)

(Total 11 marks)

Q6.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - 7x - 6$.

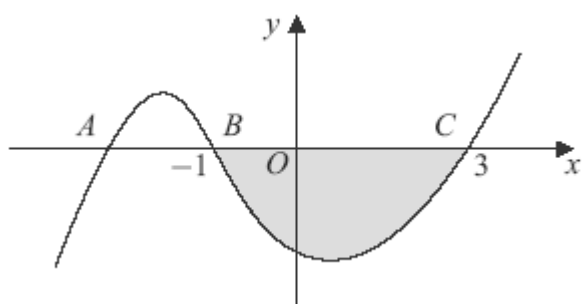
- (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

- (ii) Express $p(x) = x^3 - 7x - 6$ as the product of three linear factors.

(3)

- (b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



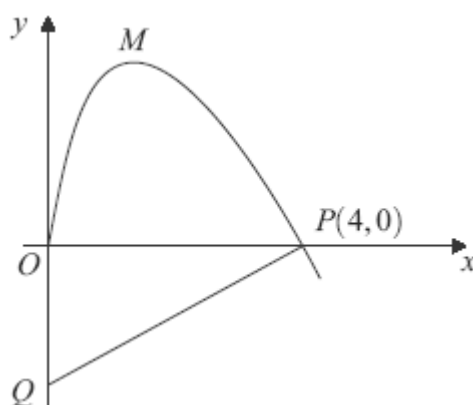
The curve cuts the x -axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

- (i) State the coordinates of the point A . (1)
 - (ii) Find $\int_{-1}^3 (x^3 - 7x - 6) dx$. (5)
 - (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x -axis between B and C . (1)
 - (iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B . (3)
 - (v) Hence find an equation of the normal to the curve at the point B . (3)
- (Total 18 marks)

Q7.

A curve, drawn from the origin O , crosses the x -axis at the point $P(4, 0)$.

The normal to the curve at P meets the y -axis at the point Q , as shown in the diagram.



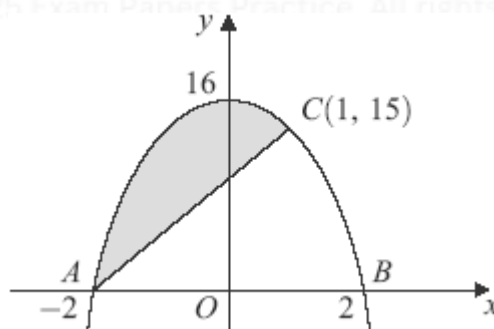
The curve, defined for $x \geq 0$, has equation

$$y = 4x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

- (a) (i) Find $\frac{dy}{dx}$. (3)
- (ii) Show that the gradient of the curve at $P(4, 0)$ is -2 . (2)
- (iii) Find an equation of the normal to the curve at $P(4, 0)$. (3)
- (iv) Find the y -coordinate of Q and hence find the area of triangle OPQ . (3)
- (v) The curve has a maximum point M . Find the x -coordinate of M . (3)
- (b) (i) Find $\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx$. (3)
- (ii) Find the total area of the region bounded by the curve and the lines PQ and QO . (3)
- (Total 20 marks)

Q8.

The curve with equation $y = 16 - x^4$ is sketched below.



The points $A(-2, 0)$, $B(2, 0)$ and $C(1, 15)$ lie on the curve.

- (a) Find an equation of the straight line AC . (3)

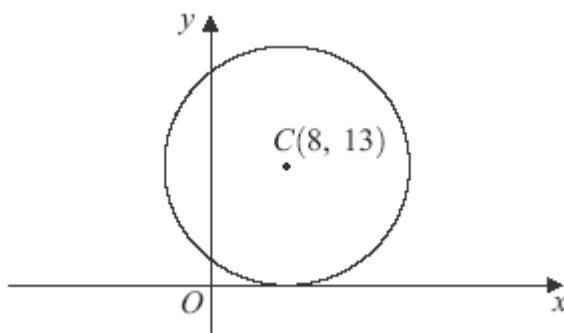
- (b) (i) Find $\int_{-2}^1 (16 - x^4) dx$. (5)

- (ii) Hence calculate the area of the shaded region bounded by the curve and the line AC . (3)

(Total 11 marks)

Q9.

The circle S has centre $C(8, 13)$ and touches the x -axis, as shown in the diagram.



- (a) Write down an equation for S , giving your answer in the form

$$(x - a)^2 + (y - b)^2 = r^2 \quad (2)$$

- (b) The point P with coordinates $(3, 1)$ lies on the circle.

- (i) Find the gradient of the straight line passing through P and C . (1)

- (ii) Hence find an equation of the tangent to the circle S at the point P , giving your answer in the form $ax + by = c$, where a , b and c are integers. (4)

- (iii) The point Q also lies on the circle S , and the length of PQ is 10. Calculate the shortest distance from C to the chord PQ . (3)

(Total 10 marks)

Q10.

- (a) Write $\sqrt{x^3}$ in the form x^k , where k is a fraction. (1)

- (b) A curve, defined for $x \geq 0$, has equation

$$y = x^2 - \sqrt{x^3}$$

(i) Find $\frac{dy}{dx}$.

(3)

- (ii) Find the equation of the tangent to the curve at the point where $x = 4$, giving your answer in the form $y = mx + c$.

(5)

(Total 9 marks)

Q11.

A curve is defined, for $t \neq 0$, by the parametric equations

$$x = 4t + 3, \quad y = \frac{1}{2t} - 1$$

At the point P on the curve, $t = \frac{1}{2}$.

- (a) Find the gradient of the curve at the point P .

(4)

- (b) Find an equation of the normal to the curve at the point P .

(3)

- (c) Find a cartesian equation of the curve.

(3)

(Total 10 marks)

Q12.

The line AB has equation $3x + 5y = 8$ and the point A has coordinates $(6, -2)$.

- (a) (i) Find the gradient of AB .

(2)

- (ii) Hence find an equation of the straight line which is perpendicular to AB and which passes through A .

(3)

- (b) The line AB intersects the line with equation $2x + 3y = 3$ at the point B . Find the coordinates of B .

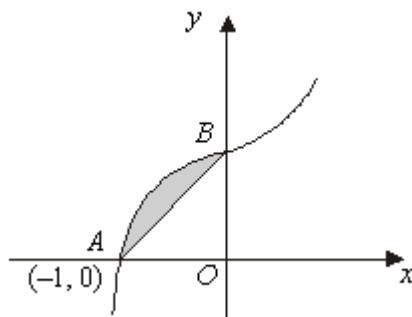
(3)

- (c) The point C has coordinates $(2, k)$ and the distance from A to C is 5. Find the **two** possible values of the constant k .

(3)
(Total 11 marks)

Q13.

The curve with equation $y = 3x^5 + 2x + 5$ is sketched below.



The curve cuts the x -axis at the point $A(-1, 0)$ and cuts the y -axis at the point B .

- (a) (i) State the coordinates of the point B and hence find the area of the triangle AOB , where O is the origin.

(3)

- (ii) Find $\int (3x^5 + 2x + 5) dx$

(3)

- (iii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

- (b) (i) Find the gradient of the curve with equation $y = 3x^5 + 2x + 5$ at the point $A(-1, 0)$.

(3)

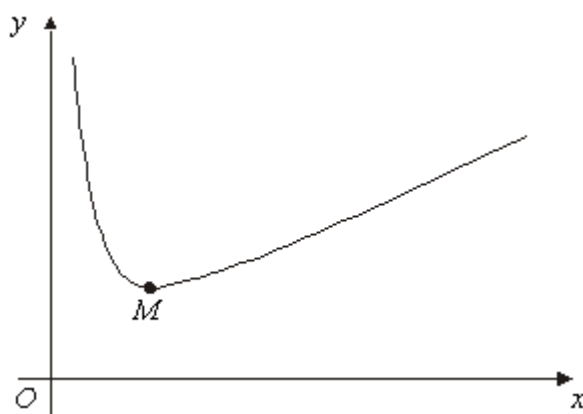
- (ii) Hence find an equation of the tangent to the curve at the point A .

(1)

(Total 14 marks)

Q14.

A curve C is defined for $x > 0$ by the equation $y = x + 1 + \frac{4}{x^2}$ and is sketched below.



- (a) (i) Given that $y = x + 1 + \frac{4}{x^2}$, find $\frac{dy}{dx}$. (3)
- (ii) The curve C has a minimum point M . Find the coordinates of M . (4)
- (iii) Find an equation of the normal to C at the point $(1, 6)$. (4)
- (b) (i) Find $\int \left(x + 1 + \frac{4}{x^2} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the lines $x = 1$ and $x = 4$ and the x -axis. (2)

(Total 16 marks)

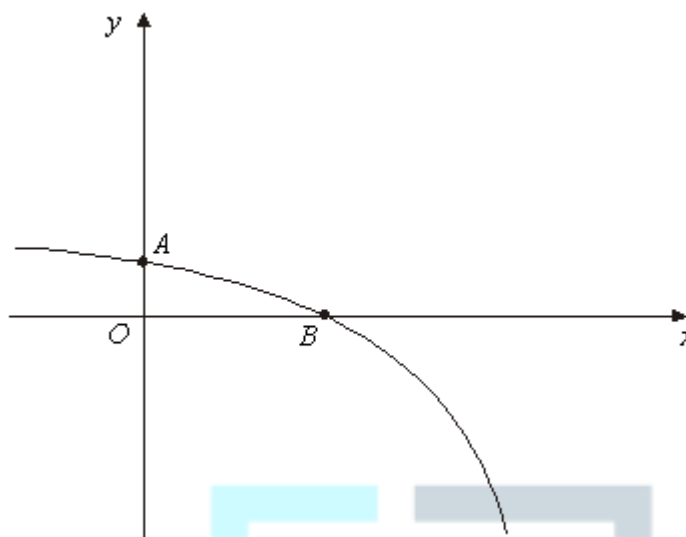
Q15.

- (a) Find $\frac{dy}{dx}$ when:
- (i) $y = (4x^2 + 3x + 2)^{10}$; (2)
- (ii) $y = x^2 \tan x$. (2)
- (b) (i) Find $\frac{dx}{dy}$ when $x = 2y^3 + \ln y$. (1)
- (ii) Hence find an equation of the tangent to the curve $x = 2y^3 + \ln y$ at the point $(2, 1)$.

(3)
(Total 8 marks)

Q16.

The sketch shows the graph of $y = 4 - e^{2x}$. The curve crosses the y -axis at the point A and the x -axis at the point B .



- (a) (i) Find $\int (4 - e^{2x}) dx$. (2)
- (ii) Hence show that $\int_0^{\ln 2} (4 - e^{2x}) dx = 4 \ln 2 - \frac{3}{2}$. (2)
- (b) (i) Write down the y -coordinate of A . (1)
- (ii) Show that $x = \ln 2$ at B . (2)
- (c) Find the equation of the normal to the curve $y = 4 - e^{2x}$ at the point B . (4)
- (d) Find the area of the region enclosed by the curve $y = 4 - e^{2x}$, the normal to the curve at B and the y -axis. (3)

(Total 14 marks)

Q17.

A curve is defined by the parametric equations

$$x = 1 + 2t, \quad y = 1 - 4t^2$$

- (a) (i) Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

(2)

- (ii) Hence find $\frac{dy}{dx}$ in terms of t .

(2)

- (b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

- (c) Find a cartesian equation of the curve.

(3)

(Total 11 marks)

Q18.

The variables x and y are related by an equation of the form

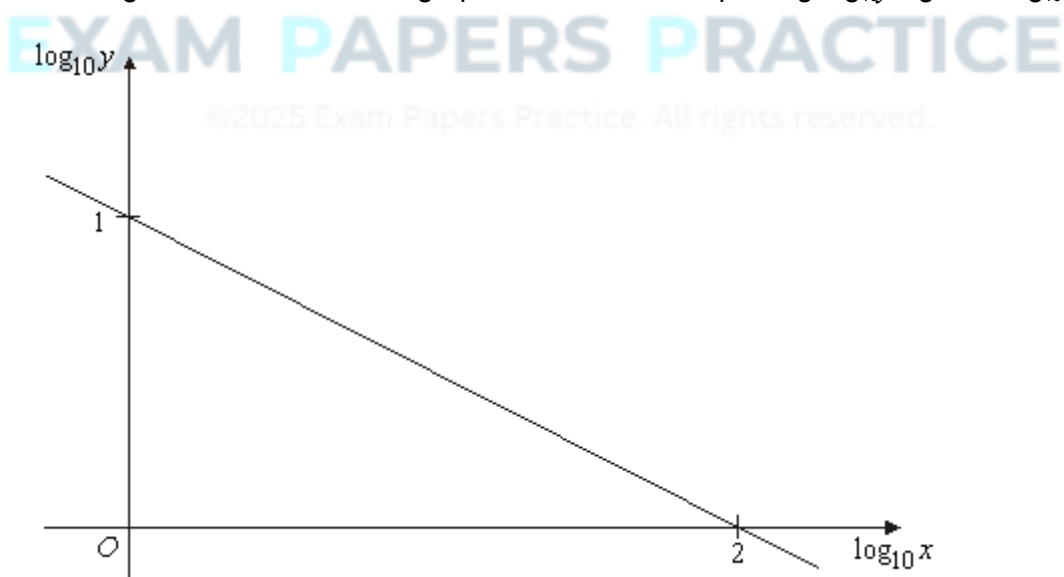
$$y = ax^b$$

where a and b are constants.

- (a) Using logarithms to base 10, reduce the relation $y = ax^b$ to a linear law connecting $\log_{10}x$ and $\log_{10}y$.

(2)

- (b) The diagram shows the linear graph that results from plotting $\log_{10}y$ against $\log_{10}x$.



Find the values of a and b .

(4)

(Total 6 marks)

Q19.

The points A and B have coordinates $(6, -1)$ and $(2, 5)$ respectively.

- (a) (i) Show that the gradient of AB is $-\frac{3}{2}$ (2)
- (ii) Hence find an equation of the line AB , giving your answer in the form $ax + by = c$, where a , b and c are integers. (2)
- (b) (i) Find an equation of the line which passes through B and which is perpendicular to the line AB . (2)
- (ii) The point C has coordinates $(k, 7)$ and angle ABC is a right angle. Find the value of the constant k . (2)

(Total 8 marks)

Q20.

A circle with centre C has equation $(x + 3)^2 + (y - 2)^2 = 25$.

- (a) Write down:
- (i) the coordinates of C ; (2)
- (ii) the radius of the circle. (1)
- (b) (i) Verify that the point $N(0, -2)$ lies on the circle. (1)
- (ii) Sketch the circle. (2)
- (iii) Find an equation of the normal to the circle at the point N . (3)
- (c) The point P has coordinates $(2, 6)$.
- (i) Find the distance PC , leaving your answer in surd form. (2)

- (ii) Find the length of a tangent drawn from P to the circle.

(3)

(Total 14 marks)

Q21.

A curve is defined for $x > 0$ by the equation

$$y = \left(1 + \frac{2}{x}\right)^2$$

The point P lies on the curve where $x = 2$.

- (a) Find the y -coordinate of P .

(1)

- (b) Expand $\left(1 + \frac{2}{x}\right)^2$.

(2)

- (c) Find $\frac{dy}{dx}$

(3)

- (d) Hence show that the gradient of the curve at P is -2 .

(2)

- (e) Find the equation of the normal to the curve at P , giving your answer in the form $x + by + c = 0$, where b and c are integers.

(4)

(Total 12 marks)

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Q22.

The point A has coordinates $(1, 1)$ and the point B has coordinates $(5, k)$.

The line AB has equation $3x + 4y = 7$.

- (a) (i) Show that $k = -2$.

(1)

- (ii) Hence find the coordinates of the mid-point of AB .

(2)

- (b) Find the gradient of AB .

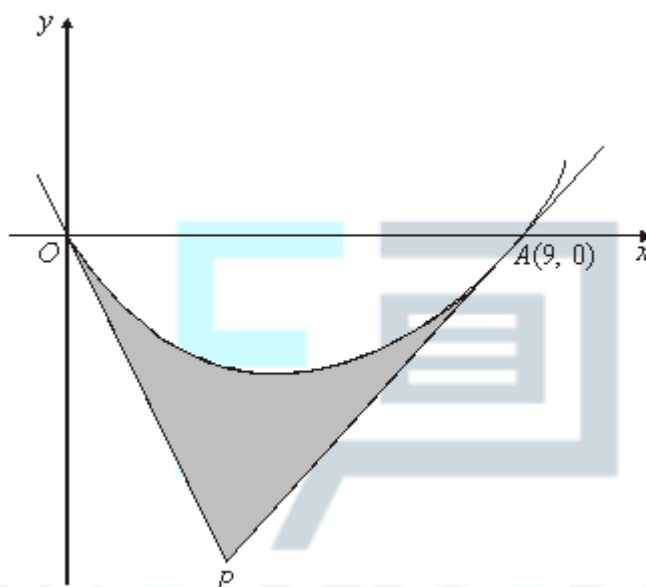
(2)

- (c) The line AC is perpendicular to the line AB .

- (i) Find the gradient of AC . (2)
 - (ii) Hence find an equation of the line AC . (1)
 - (iii) Given that the point C lies on the x -axis, find its x -coordinate. (2)
- (Total 10 marks)

Q23.

A curve, drawn from the origin O , crosses the x -axis at the point $A(9, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = x^{\frac{3}{2}} - 3x$$

- (a) Find $\frac{dy}{dx}$. (2)
- (b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O . (2)
- (ii) Show that the equation of the tangent at $A(9, 0)$ is $2y = 3x - 27$. (3)

(iii) Hence find the coordinates of the point P where the two tangents meet.

(3)

(c) Find $\int \left(x^{\frac{3}{2}} - 3x \right) dx$.

(3)

(d) Calculate the area of the shaded region bounded by the curve and the tangents OP and AP .

(5)

(Total 18 marks)

Q24.

A curve is defined by the parametric equations

$$x = 3 - 4t \quad y = 1 + \frac{2}{t}$$

(a) Find $\frac{dy}{dx}$ in terms of t .

(4)

(b) Find the equation of the tangent to the curve at the point where $t = 2$, giving your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(4)

(c) Verify that the cartesian equation of the curve can be written as

$$(x - 3)(y - 1) + 8 = 0$$

(3)

(Total 11 marks)

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Q25.

The point A has coordinates $(1, 7)$ and the point B has coordinates $(5, 1)$.

(a) (i) Find the gradient of the line AB .

(2)

(ii) Hence, or otherwise, show that the line AB has equation $3x + 2y = 17$.

(2)

(b) The line AB intersects the line with equation $x - 4y = 8$ at the point C . Find the coordinates of C .

(3)

(c) Find an equation of the line through A which is perpendicular to AB .

(3)
(Total 10 marks)

Q26.

A curve has equation $y = 7 - 2x^5$.

- (a) Find $\frac{dy}{dx}$. (2)
- (b) Find an equation for the tangent to the curve at the point where $x = 1$. (3)
- (c) Determine whether y is increasing or decreasing when $x = -2$. (2)
- (Total 7 marks)

Q27.

The point A has coordinates $(11, 2)$ and the point B has coordinates $(-1, -1)$.

- (a) (i) Find the gradient of AB . (2)
- (ii) Hence, or otherwise, show that the line AB has equation $x - 4y = 3$. (2)
- (b) The line with equation $3x + 5y = 26$ intersects the line AB at the point C . Find the coordinates of C . (3)
- (Total 7 marks)

Q28.

A curve has equation $y = x^5 - 6x^3 - 3x + 25$.

- (a) Find $\frac{dy}{dx}$. (3)
- (b) The point P on the curve has coordinates $(2, 3)$.
- (i) Show that the gradient of the curve at P is 5. (2)
- (ii) Hence find an equation of the normal to the curve at P , expressing your

answer in the form $ax + by = c$, where a , b and c are integers.

(3)

- (c) Determine whether y is increasing or decreasing when $x = 1$.

(2)

(Total 10 marks)

Q29.

A curve is defined for $x > 0$ by the equation $y = x + \frac{2}{x}$.

- (a) (i) Find $\frac{dy}{dx}$.

(3)

- (ii) Hence show that the gradient of the curve at the point P where $x = 2$ is $\frac{1}{2}$.

(1)

- (b) Find an equation of the normal to the curve at this point P .

(4)

(Total 8 marks)

Q30.

The point A has coordinates $(6, 5)$ and the point B has coordinates $(2, -1)$.

- (a) Find the coordinates of the midpoint of AB .

(2)

- (b) Show that AB has length $k\sqrt{13}$, where k is an integer.

(3)

- (c) (i) Find the gradient of the line AB .

(2)

- (ii) Hence, or otherwise, show that the line AB has equation $3x - 2y = 8$.

(2)

- (d) The line AB intersects the line with equation $2x + y = 10$ at the point C . Find the coordinates of C .

(3)

(Total 12 marks)

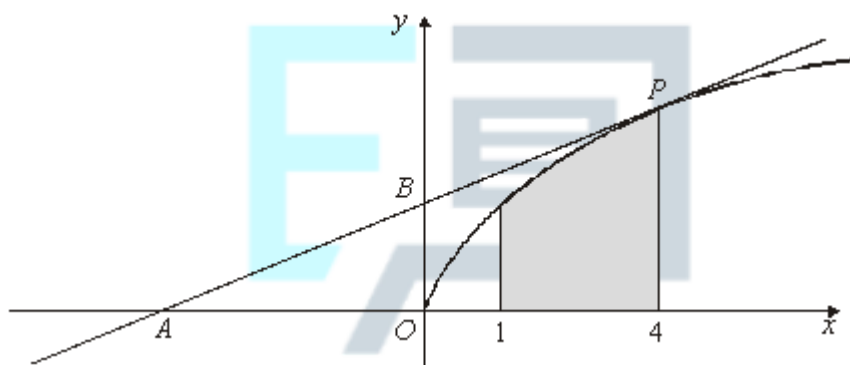
Q31.

A circle has centre $C(2, -1)$ and radius 5. The point P has coordinates $(6, 2)$.

- (a) Write down an equation of the circle. (3)
- (b) Verify that the point P lies on the circle. (2)
- (c) Find the gradient of the line CP . (2)
- (d) (i) Find the gradient of a line which is perpendicular to CP . (2)
- (ii) Hence find an equation for the tangent to the circle at the point P . (1)
- (Total 10 marks)

Q32.

The diagram shows a curve C with equation $y = \sqrt{x}$. The point O is the origin $(0, 0)$.



The region bounded by the curve C , the x -axis and the vertical lines $x = 1$ and $x = 4$ is shown shaded in the diagram.

- (a) (i) Write $\sqrt{x}$ in the form x^p , where p is a constant. (1)
- (ii) Find $\int \sqrt{x} \, dx$. (2)
- (iii) Hence find the area of the shaded region. (3)
- (b) The point on C for which $x = 4$ is P . The tangent to C at the point P intersects the x -axis and the y -axis at the points A and B respectively.
- (i) Find an equation for the tangent to the curve C at the point P . (4)

- (ii) Find the area of the triangle AOB . (3)
- (c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C . (2)
- (d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1} \, dx$, giving your answer to three significant figures. (4)
- (Total 19 marks)**



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